Yashwanth Venkat

+1(910)-479-4815 | yashwanthg2111@gmail.com | http://www.linkedin.com/in/yashwanth-venkat-g-1b5828173

PROFESSIONAL SUMMARY:

Overall 7+ years of experience in Data engineering and skilled in designing, developing, and optimizing high-scale, mission-critical data ecosystems, with expertise in Big Data Engineering, Cloud Data Platforms, and Advanced Data Pipelines across multi-cloud and on-premises environments.

- Extensive experience with **Hadoop, Apache Spark, Databricks, and Snowflake**, optimizing petabyte-scale distributed data processing workloads, leveraging Spark tuning, parallelization strategies, and advanced memory management techniques
- Aptitude in **EMR, EC2, S3, Lambda, Athena, Glue, RDS, Redshift,** and **DynamoDB**, building serverless big data architectures and auto-scalable processing pipelines.
- Proficient in **Azure Data Lake Storage, Azure Data Factory, Synapse**, and **Azure SQL Server,** designing high-performance ELT/ETL solutions leveraging **PolyBase**, **Dataflows**, and **Spark pools**.
- Hands-on experience with **Apache Beam, DataProc, BigQuery, Dataflow**, and **Composer**, developing low-latency, streaming analytics solutions with real-time event processing.
- Well-Verse in **Python, Scala, and Java,** developing highly parallelized ETL/ELT frameworks, building custom ML-based anomaly detection models, and optimizing multi-threaded processing logic in **Spark** and **Databricks**.
- Demonstrated Mastery of in **SQL, PL/SQL, Hive, MySQL, DB2, PostgreSQL**, and **Oracle**, optimizing terabyte-scale queries, implementing advanced indexing, partitioning, materialized views, and column store strategies for real-time, high-performance querying.
- Skillset experience in version control using **Git, GitLab, GitHub, SVN,** and **CVS**, implementing best-practice branching strategies like **GitFlow** and **Trunk-based development**, while designing containerized big data pipelines with **Docker** and **Kubernetes** and automating CI/CD workflows using **Jenkins**, **GitHub Actions**, and **Terraform** for seamless multi-cloud deployments.
- Designed and implemented ML-driven data pipelines, integrating **Scikit-learn**, **TensorFlow**, and **PyTorch** into big data ecosystems, automating anomaly detection, forecasting, and entity resolution at scale.
- Proficient in **Apache Airflow**, **Oozie**, **Azkaban**, and **AWS Step Functions**, designing dynamic DAG-based orchestration workflows with intelligent retry mechanisms and SLA-driven automation.
- Expertise in management, deployment, and optimization of **Hadoop** clusters built on **Cloudera** and **Hortonworks**, enhancing **HDFS/YARN** configurations, ensuring high-availability setups, and implementing robust **Kerberos** security protocols Architected and implemented multi-layered data architectures with **Kimball** & **Inmon** methodologies, utilizing **Erwin** for data modeling, and optimizing schema evolution strategies across cloud data warehouses.
- Deep expertise in multi-format data processing, working extensively with **CSV**, **Parquet**, **ORC**, **Avro**, **JSON**, **XML**, and raw text across distributed computing environments, optimizing serialization & compression strategies.
- Experience in designing and developing the executive-level, real-time dashboards using **Tableau**, **Power BI**, and **Looker**, integrating multi-source analytics pipelines, embedding custom **DAX/MDX** calculations for business intelligence.
- Expert in constructing the high-performance OLAP cubes using **Kyvos, Apache Druid, Azure Analysis Services,** and **SSAS**, supporting multi-dimensional, ad-hoc analytics for enterprise-scale reporting.
- Mastery in **JIRA**, **ServiceNow**, **Rally** and **TFS**, implementing agile sprint planning, backlog grooming, and DevOps-driven incident management for high-availability systems.

TECHINCAL SKILLS:

Bigdata Frameworks	Apache Spark, Hive, Sqoop, Kafka, Hadoop, MapReduce, HDFS, Pig, Oozie, Apache Airflow
Languages	Python, Shell Scripting, PySpark, HiveQL, Scala, Java
Visualization	Power BI, Tableau, Looker
Data Base	Microsoft SQL Server, PostgreSQL, MySQL, Oracle, DB2
Datawarehouse	Snowflake, Teradata
Cloud	AZURE: Azure: Azure Data Lake, Azure Data Factory, Azure Databricks, Azure SQL Database, Azure Synapse, Azure Analysis Services, Azure Blob Storage, Azure Monitor, Azure key vault, Azure HDinside, Azure Data Explorer. AWS: AWS S3, EC2, Redshift, Redis, EMR, Kinesis, Lambda, Step Functions, Glue, Athena, DynamoDB, Elasticsearch, Service Catalog, CloudWatch, IAM, SageMaker, RDS, API Gateway, AWS CloudWatch, Aws Glue ML Transforms, Amazon Athena GCP: Google Cloud Storage, Dataflow, BigQuery, DataProc, Pub/Sub, Composer, AI Platform, Cloud Functions, Dataplex, Data Catalog.
IDE's	PyCharm, Jupyter Notebooks, Visual Studio Code, IntelliJ IDEA, Eclipse
Web programming	HTML, CSS, JavaScript, Angular, Flask, Django
Workflow Automation	Apache Airflow, Oozie, Step Functions, Prefect, Azkaban, Apache NiFi
Security & Access Control	IAM, OAuth, JWT, Apache Ranger, Kerberos, SSL/TLS Encryption
Machine Learning & AI	Scikit-learn, TensorFlow, PyTorch, MLlib, SageMaker, AutoML

PROFESSIONAL EXPERIENCE:

Freddie Mac **May 2023 - Present**
Senior Data Engineer
Responsibilities:

- Architected and engineered highly scalable ETL/ELT data pipelines integrating diverse data sources such as **SQL Server, Oracle, web analytics, APIs** to enable real-time insights and analytics at enterprise scale.
- Designed and optimized high-performance **Databricks** architectures and utilizing **Spark**'s distributed computing capabilities to process petabyte-scale datasets efficiently and optimizing the query execution time.
- Automated complex workflows in **Azure Databricks**, Arranging **SQL** and **Python**-based transformations with **Databricks Jobs**, **Delta Live Tables,** and **Apache Airflow,** ensuring fault-tolerant data processing pipelines.
- Configured **Azure Data Lake Gen2** and Develop and maintain **RESTful APIs** to storage for high-throughput of Data streaming, managing structured and semi-structured datasets such as **CSV, Parquet, Delta formats**, Managing storage costs and access speeds for high-frequency analytics workloads.
- Engineered and optimized **Oracle SQL** and **PL/SQL** procedures, implementing parallel execution strategies and query tuning techniques and using **DBMS** Scheduler for automating batch processing.
- Pioneered in the migration of mission-critical enterprise data warehouse from **Oracle** to **Azure SQL Data Warehouse**, implementing distributed processing, column store indexing, and partitioning strategies to support near real-time dashboarding.
- Designed real-time streaming architectures using **Azure Event Hubs, Kafka**, and **Spark Structured Streaming**, enabling near-instantaneous processing of clickstream and customer engagement data.

- Developed highly interactive, self-service **Tableau** dashboards for C-level executives, integrating data from **Google BigQuery, Azure SQL Data Warehouse**, and on-premises databases and empowering data-driven business decisions.
- Executed outstanding performance of **ETL** workflows, identifying bottlenecks and optimizing cluster configurations in Databricks, tuning Spark Shuffle, caching, partitioning, and DAG optimizations, resulting in a reduction in processing time.
- Built an Alteryx-based data validation framework to automate multi-system for Cross-checking processes between disparate datasets, flagging inconsistencies and ensuring high data quality, publishing insights to **Power BI** Server.
- Enabled multi-cloud data strategies, integrating **Azure, Google BigQuery**, and third-party **API**-based data sources into a unified data warehouse while implementing cost-efficient cloud storage and query execution strategies.
- Designed, developed, and optimized data structures, SQL queries, and automated pipelines for migrating legacy systems to **Snowflake Data Warehouse**, integrating **Airflow, AWS S3, Web APIs, and Teradata** for seamless data processing and reporting.
- Designed and enforced data governance, security and compliance policies across **cloud data lakes** for ensuring adherence to **GDPR, CCPA**, and enterprise security standards for sensitive data handling.
- Engineered high-performance **OLAP** cubes in **Azure Analysis Services**, optimizing complex aggregations for enterprise reporting, ensuring real-time insights for sales, marketing, and finance teams.
- Utilized **J2EE** tools to build and maintain enterprise-grade web applications, ensuring seamless integration with backend data processing pipelines and enhancing the scalability and performance of data-driven services.
- Developed and published custom Python packages to **PyPI**, ensuring seamless distribution and easy installation across teams and projects. Implemented version control for packages, facilitating collaborative development, promoting code reusability, and minimizing duplicated efforts by enabling consistent access to shared functionalities

Environment: SQL Server, Oracle, PL/SQL, Google BigQuery, RESTful APIs ,Apache Spark, Delta Lake, Databricks, Azure Data Factory, Apache Airflow, Azure Event Hubs, Kafka, Azure SQL Data Warehouse, Azure Analysis Services, Alteryx, Power BI Server, CI/CD pipelines , Snowflake, Teradata.

Fidelity Investments **Jan21 – Dec 2022**
Senior AWS Data Engineer
Responsibilities:

- Architected and optimized large-scale distributed data processing workflows using **Apache Spark**, **Hadoop**, and **Hive** integrated with **AWS S3** for scalable storage, handling terabytes of structured and unstructured data. This infrastructure was optimized with **EMR** to improve processing efficiency in an enterprise environment.
- Developed highly optimized Spark applications using **PySpark**, **DataFrames**, **RDDs**, and **SparkSQL**, tuning performance with strategies such as broadcast joins, partitioning, and caching. Utilized **AWS Glue** for efficient ETL processes, reducing query execution times.
- Engineered robust ETL pipelines, integrating real-time and batch data ingestion from multiple data sources using **Kafka**, **Sqoop**, **AWS Lambda**, and **Spark Streaming** to ensure reliable data flow into **AWS S3**, **Hive**, and **AWS Redshift**.
- Designed and implemented schema evolution strategies using **Hive external tables**, **partitioning**, **bucketing**, and format optimizations with **ORC**/**Parquet**. This complement is used by **AWS Athena** to query **S3-**stored data for faster BI and reporting workloads.
- Developed advanced data validation and enrichment frameworks in **PySpark** using custom **UDFs** and **AWS Glue ML Transforms** to incorporate machine learning-based anomaly detection to enhance data quality and reduce inconsistencies across multiple data sources.
- Created custom data processing pipelines to extract, transform, and load high-velocity consumer response data into **AWS S3** and **Hive**, optimizing storage and query execution strategies to support real-time and batch analytical workloads. Utilized **AWS Redshift** for further data aggregation and reporting.

- Organized and automated complex workflows using **AWS Step Functions** and **Apache Airflow**, building and scheduling dynamic DAGs that adapt to dependencies, ensuring fault tolerance, retry mechanisms, and SLA-driven data processing.
- Designed optimized data extraction strategies from legacy systems with **Sqoop** and **AWS RDS**, improving ingestion speeds through direct JDBC optimizations and split-by tuning. Also, **AWS DynamoDB** utilized for low-latency NoSQL storage.
- Developed dynamic Python-based ETL frameworks to adapt to schema changes, utilizing **pandas**, **NumPy**, and **Spark DataFrames** for high-speed data cleansing, transformation and aggregation. Utilized **Elasticsearch** for fast search queries on processed datasets.
- Implemented complex Hive queries involving multi-level joins, window functions, and dynamic partition pruning. These optimizations enabled real-time analytics, with **AWS Athena** used for querying large datasets stored in **S3**, drastically reducing reporting times for business stakeholders.
- Devised a metadata-driven ingestion pipeline to auto-discover and register datasets in **AWS Glue** and **Hive**, enabling self-service analytics. This approach reduced manual data engineering efforts and improved scalability of data cataloging.
- Optimized cluster configurations and resource allocation in **Spark** and **Hadoop** on **AWS EMR**, fine-tuning **YARN** resource management, driver/executor tuning, and shuffle optimizations. This resulted in a reduction in infrastructure costs, and real-time analytics were powered using **Kinesis** and **AWS Lambda**.
- Built Alteryx-based data validation frameworks, automating multi-system reconciliation processes between disparate datasets, flagging inconsistencies, and ensuring high data quality, publishing insights to **Tableau** Server

Environment: PySpark, Hadoop, Hive, Spark Streaming, Kafka, Airflow, HDFS, ORC/Parquet, Sqoop, Hive UDFs, MySQL, RDBMS, Linux, Python, YARN, Apache Ranger, Databricks, MLlib,Snowflake and Teradata, AWS S3, EC2, Redshift, Redis, EMR, Kinesis, Lambda, Step Functions, Glue, Athena, DynamoDB, Elasticsearch, Service Catalog, CloudWatch, IAM, SageMaker, RDS, API Gateway, AWS CloudWatch, AWS snowflake, Tableau.

Capital One **Jan 2019 – Jan 2021**
Big data engineer
Responsibilities:

- Build and deployed multi-node **Hadoop** clusters from scratch, configuring **HDFS, YARN, MapReduce, Hive** and **HBase** for high-performance distributed data processing.
- Developed and optimized highly parallelized **Java-based MapReduc**e programs, implementing custom partitioners, combiners, and in-memory caching strategies to significantly reduce computation time for complex data cleaning and preprocessing pipelines.
- Designed and implemented **Hadoop**-based **ETL** pipelines, utilizing **Hive**, **Pig**, and **Sqoop** to efficiently transform and load terabyte-scale datasets between **HDFS**, **relational databases**, and **cloud storage platforms**.
- Supervised **Hadoop** ecosystem performance tuning, optimizing block size, memory allocation, compression strategies such as **Snappy, LZO**, and file formats like **ORC, Parquet** reducing query execution times.
- Engineered a distributed, fault-tolerant data ingestion framework, automating data imports and exports between **HDFS, Hive**, and **RDBMS** using **Sqoop, Kafka, and Spark**, ensuring seamless data movement across enterprise systems.
- Designed custom **Hive UDF**s in **Java**, extending **Hive**'s capabilities for advanced text processing, fuzzy matching, and analytics at scale.
- Managed **Hadoop** job workflows using **Apache Oozie** and **Airflow**, automating complex dependencies, job retries, and error handling mechanisms for critical data processing tasks.
- Configured and managed Cloudera-based **Hadoop** infrastructure, overseeing multi-node cluster deployments, upgrades, access control policies like **Kerberos, Ranger**, and high-availability setups, ensuring system uptime.

- Developed and optimized **HBase** schema designs, implementing row key modeling, Bloom filters, and coprocessors to support millisecond-level random access queries on billions of records.
- Utilized **Zookeeper** for distributed coordination and leader election mechanisms, ensuring cluster stability and consistent metadata synchronization across services like **HBase, Kafka,** and **Oozie**.
- Designed near real-time data processing pipelines using **Kafka , Spark** Streaming, enabling low-latency analytics for clickstream, IoT, and log processing Use cases.
- Implemented advanced cluster monitoring and logging solutions, integrating **Grafana**, **Prometheus** and **ELK Stack** to proactively detect performance bottlenecks and system failures.
- Managed data governance and compliance initiatives, implementing fine-grained access control such as **Apache Range**r, **Sentry** and encryption strategies to ensure security for sensitive and **PII data**.
- Built self-service analytical environments for business users, integrating **Hive, Presto**, and **BI tools** to enable ad-hoc querying of massive datasets.

Environment: Java, Python, SQL, Hadoop HDFS, Hadoop YARN, Hadoop MapReduce ,Apache Spark, Apache Hive, Apache Pig, Apache HBase, Apache Kafka, Apache Sqoop, Apache Oozie, Apache Airflow, Cloudera, Kerberos, Apache Ranger, Apache Sentry, Apache Zookeeper, ORC, Parquet, Snappy, LZO, Grafana, Prometheus, ELK Stack, ELK Stack Elasticsearch, ELK Stack Logstash, ELK Stack Kibana, AWS S3,AWS Athena, AWS EC2,AWS Auto-scaling , Azure, Presto, Apache Hive, AWS Athena, Tableau**.**

Tata Consultancy Services — Jan 2017 – Jan 2019

Python Developer
Responsibilities:

- Involved in the full project lifecycle from design, development to deployment, testing, implementation, and post-production support, ensuring seamless application delivery and system performance.
- Developed web-based applications using **PHP, XML, JSON**, and **MVC3** architecture using **Python scripts** to automate database updates, content management, and file manipulations for enhanced operational efficiency.
- Built robust database models, **API**s, and views using **Python** and facilitating smooth integration then after ensuring the creation of an interactive and efficient **web application**.
- Proficiently built and developed the presentation layer of web applications, utilizing **HTML, CSS, JavaScript, JQuery, AJAX**, and **Bootstrap** to build responsive, dynamic, and user-friendly interfaces.
- Utilized **PyPI** to automate various Python dependencies, optimizing package installations and ensuring compatibility across diverse environments which improves project efficiency and reduces setup time.
- Developed **XML** schema documents and implemented frameworks for parsing **XML** data, ensuring the successful exchange of structured data between systems and applications.
- Utilized **Python** for processing **XML** and **JSON** data, implementing core business logic and enhancing data exchange, contributing to the overall automation and efficiency of the web application.
- Managed large datasets using **Pandas data frames** and **MySQL**, integrating with multiple relational databases like **MySQL, Oracle, PostgreSQL** and contributing to the development of REST **API**s with **Flask, SQL Alchemy,** and **OpenStack API integration**, while automating deployment using **Jenkins** and **Git** within **Agile SCRUM** methodologies.

Environment: Python, XML, MySQL, Apache, HTML, CSS, JavaScript, Shell Scripts, Oracle, PostgreSQL, REST, SOAP, JSON, Django.